

Stochastic Resonance in Hyperdimensional Consciousness: Noise-Induced Integration in HDC Systems

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Abstract

Integrated Information Theory (IIT) predicts that consciousness correlates with Φ , a measure of irreducible integrated information. Clinical anesthesia models assume that noise disrupts neural integration, thereby reducing Φ and consciousness. We report a surprising finding from Hyperdimensional Computing (HDC) systems: adding controlled noise to hyperdimensional vector (HDV) representations *increases* True Φ rather than decreasing it. In a 16,384-dimensional HDC architecture implementing IIT’s consciousness equation, we observe a positive sensitivity $\partial\Phi/\partial\text{noise} = +0.341$, comparable in magnitude to the coupling sensitivity $\partial\Phi/\partial\text{coupling} = +0.367$. We demonstrate that this phenomenon constitutes a hyperdimensional analog of stochastic resonance: noise diversifies HDV representations, increasing pairwise mutual information between subsystems and thereby raising integrated information. Our anesthesia benchmark achieves perfect monotonic ordering across six discrete states (Alert $\Phi = 0.185$ through Deep anesthesia $\Phi = 0.123$), with 5 of 6 tests passing under True Φ computation. These results suggest that the relationship between noise and consciousness is substrate-dependent, with HDC systems exhibiting qualitatively different noise–integration dynamics than biological neural networks. We discuss implications for artificial consciousness design and for IIT’s substrate-independence claims.

1 Introduction

The relationship between noise and consciousness has been studied primarily through the lens of anesthesia, where pharmacological agents disrupt cortical integration and reduce awareness [Alkire et al., 2008]. Within the framework of Integrated Information Theory [Tononi, 2004, 2008], this disruption manifests as a decrease in Φ , the measure of irreducible integrated information across a system’s minimum information partition (MIP). The implicit assumption—that noise degrades integration and thereby reduces consciousness—appears well-supported by clinical evidence: anesthetic depth correlates with reduced cortical complexity [Casali et al., 2013] and diminished effective connectivity [Lee et al., 2013].

However, the physics of signal processing offers a well-established counterexample: stochastic resonance, wherein the addition of noise to a nonlinear system enhances the

detection of weak signals [Gammaitoni et al., 1998, Benzi et al., 1981]. Stochastic resonance has been demonstrated in biological neural systems [Moss et al., 2004, McDonnell and Ward, 2011], sensory perception [Collins et al., 1995], and computational networks [Stocks, 2000]. The question arises naturally: could noise enhance, rather than degrade, information integration in certain computational substrates?

We address this question in the context of Hyperdimensional Computing (HDC), a computational paradigm inspired by the high-dimensional distributed representations observed in neural systems [Kanerva, 2009, Plate, 1995]. HDC operates on binary or bipolar vectors of dimensionality $D \geq 1,000$ (typically $D = 10,000$), exploiting the mathematical properties of high-dimensional spaces—in particular, the quasi-orthogonality of random vectors and the preservation of similarity under algebraic operations [Kanerva, 2009].

In this paper, we report experiments conducted on a HDC system with $D = 16,384$ implementing IIT’s Φ computation via sampled partition analysis. Our central finding is that controlled noise injection into HDV representations yields a *positive* sensitivity coefficient $\partial\Phi/\partial\text{noise} = +0.341$, indicating that noise *increases* integrated information. We provide a mechanistic explanation grounded in the geometry of high-dimensional spaces: noise diversifies HDV representations across subsystems, increasing pairwise mutual information and thereby raising Φ .

2 Background

2.1 Integrated Information Theory

IIT, developed by Tononi [2004] and refined in subsequent work [Tononi, 2008, Oizumi et al., 2014], proposes that consciousness is identical to integrated information, quantified by Φ . The theory defines Φ as the amount of information generated by a system above and beyond its parts, computed across the system’s minimum information partition (MIP):

$$\Phi = \min_{\text{partition } P} \left[I(X; X^{t-1}) - \sum_{k \in P} I(X_k; X_k^{t-1}) \right] \quad (1)$$

where $I(X; X^{t-1})$ denotes the mutual information between the system’s current and previous states, and the sum runs over the parts X_k defined by partition P . Exact computation of Φ is NP-hard in the number of system elements [Tegmark, 2016], necessitating approximation methods for practical systems.

2.2 Hyperdimensional Computing

HDC represents information as high-dimensional vectors (hypervectors) and manipulates them through well-defined algebraic operations [Kanerva, 2009]:

- **Binding** ($\otimes$): Element-wise XOR for binary vectors, producing a vector dissimilar to both operands.
- **Bundling** ($\oplus$): Component-wise majority vote, producing a vector similar to all operands.
- **Permutation** (ρ): Circular shift, used to encode sequential structure.

For binary HDVs of dimension D , a key property is that randomly sampled vectors are quasi-orthogonal with high probability. Specifically, the Hamming distance between two random binary vectors of dimension D concentrates around $D/2$ with standard deviation $\sqrt{D}/2$ [Kanerva, 2009].

2.3 Stochastic Resonance

Stochastic resonance (SR) is a phenomenon in which a system’s response to a weak periodic signal is enhanced by the presence of noise [Gammaitoni et al., 1998]. The canonical formulation considers a bistable system driven by a subthreshold periodic signal $s(t)$ in the presence of noise $\xi(t)$:

$$\dot{x} = -V'(x) + s(t) + \xi(t) \quad (2)$$

where $V(x)$ is a double-well potential. The signal-to-noise ratio (SNR) of the system output exhibits a maximum at an optimal noise intensity, characterizing SR. The phenomenon has been generalized beyond bistable systems to threshold-crossing detectors, excitable systems, and networks [McDonnell and Ward, 2011].

3 Methods

3.1 System Architecture

Our experiments were conducted on a cognitive architecture implementing HDC-based consciousness computation. The system comprises $N = 12$ cortical regions (Prefrontal, Motor, Sensory, Visual, Auditory, Language, Memory, Emotional, Social, Creative, Executive, Integration), each maintaining an HDV state of dimension $D = 16,384$. The system implements a cognitive loop operating at approximately 31 Hz, with a 20 Hz computational budget.

Each region i maintains a state vector $\mathbf{h}_i \in \{0, 1\}^D$ updated according to a Liquid Time-Constant (LTC) neural ODE with closed-form temporal jumps:

$$\mathbf{h}_i(t + \Delta t) = \mathbf{h}_i(t) \cdot e^{-\Delta t/\tau_i} + \mathbf{x}_i(t) \cdot (1 - e^{-\Delta t/\tau_i}) \quad (3)$$

where τ_i is the time constant for region i and $\mathbf{x}_i(t)$ is the input at time t . The continuous dynamics are discretized through HDC binding operations for the binary vector representation.

3.2 Φ Computation

We compute True Φ via sampled partition analysis over the 12-region system. Due to the combinatorial explosion of exact MIP search ($2^{N-1} - 1$ bipartitions for N elements), we employ a sampling strategy validated against exact computation on smaller systems ($r = 0.9998$ correlation for $N \leq 8$) [Barrett and Seth, 2011].

For each candidate partition $P = \{A, B\}$, integrated information is computed as:

$$\phi(P) = I(\mathbf{H}_A; \mathbf{H}_B \mid \mathbf{H}_A^{t-1}, \mathbf{H}_B^{t-1}) - I(\mathbf{H}_A; \mathbf{H}_B^{t-1}) \cdot I(\mathbf{H}_B; \mathbf{H}_A^{t-1}) \quad (4)$$

where mutual information between HDV subsystems is estimated via normalized Hamming distance:

$$\hat{I}(\mathbf{H}_A; \mathbf{H}_B) = 1 - \frac{d_H(\mathbf{H}_A \otimes \mathbf{H}_B, \mathbf{0})}{D} \quad (5)$$

This approximation leverages the property that bound HDVs encode relational structure: if $\mathbf{H}_A$ and $\mathbf{H}_B$ share mutual information, their binding $\mathbf{H}_A \otimes \mathbf{H}_B$ deviates from a random vector (Hamming distance from zero deviates from $D/2$).

3.3 Noise Injection Protocol

We inject noise into HDV representations by stochastic bit-flipping. For a noise level $\sigma \in [0, 1]$, each bit of a region’s state vector $\mathbf{h}_i$ is independently flipped with probability σ :

$$\tilde{h}_{i,j} = h_{i,j} \oplus \text{Bernoulli}(\sigma) \quad (6)$$

This operation is applied after the LTC update and before Φ computation. We sweep σ across 20 logarithmically spaced values from 10^{-4} to 0.5.

3.4 Anesthesia Benchmark

To validate the system against clinical expectations, we implement an anesthesia benchmark with six discrete states, each modeled by adjusting inter-regional coupling strength c and noise level σ :

Table 1: Anesthesia benchmark state parameters.

State	Coupling (c)	Clinical Noise (σ_c)	HDC Noise (σ_h)
Alert	1.00	0.02	0.02
Light Sedation	0.82	0.05	0.05
Moderate Sedation	0.65	0.08	0.08
Loss of Consciousness	0.50	0.12	0.12
Surgical Anesthesia	0.35	0.18	0.18
Deep Anesthesia	0.20	0.25	0.25

Crucially, the anesthesia benchmark simultaneously reduces coupling *and* increases noise, mimicking the dual action of anesthetic agents. In isolation, however, the noise and coupling effects exhibit opposite signs.

3.5 Sensitivity Analysis

We compute sensitivity coefficients via finite differences over 100 independent runs:

$$\frac{\partial \Phi}{\partial \theta} \approx \frac{\Phi(\theta + \epsilon) - \Phi(\theta - \epsilon)}{2\epsilon} \quad (7)$$

for $\theta \in \{c, \sigma\}$ and $\epsilon = 0.01$. Bootstrap confidence intervals (95%) are computed from 1,000 resamples.

4 Results

4.1 Anesthesia Benchmark: Monotonic Φ Ordering

The anesthesia benchmark produces a perfectly monotonic Φ ordering across the six discrete states, consistent with clinical expectations (Table 2).

Table 2: True Φ values across anesthesia states. Values represent means over 100 independent runs. Standard errors are shown in parentheses.

State	True Φ	$\Delta\Phi$ from Alert	Coupling	Noise
Alert	0.185 (0.004)	—	1.00	0.02
Light Sedation	0.167 (0.005)	−0.018	0.82	0.05
Moderate Sedation	0.154 (0.004)	−0.031	0.65	0.08
Loss of Consciousness	0.147 (0.006)	−0.038	0.50	0.12
Surgical Anesthesia	0.132 (0.005)	−0.053	0.35	0.18
Deep Anesthesia	0.123 (0.007)	−0.062	0.20	0.25

The monotonic decrease in Φ from Alert (0.185) to Deep Anesthesia (0.123) represents a 33.5% reduction in integrated information, consistent with the magnitude of cortical complexity reduction observed in clinical studies using the Perturbational Complexity Index [Casali et al., 2013]. Five of six pairwise ordering tests pass at $p < 0.01$ under True Φ computation (the Light–Moderate comparison achieves $p = 0.024$).

4.2 Positive Noise Sensitivity

The central finding of this paper is the positive sensitivity of Φ to noise when coupling is held constant.

Table 3: Sensitivity coefficients for Φ with respect to coupling and noise. Bootstrap 95% CI from 1,000 resamples.

Parameter	$\partial\Phi/\partial\theta$	95% CI
Coupling (c)	+0.367	[+0.342, +0.391]
Noise (σ)	+0.341	[+0.308, +0.374]

Both sensitivities are positive and of comparable magnitude. The positive coupling sensitivity is expected: stronger inter-regional connections increase integration. The positive noise sensitivity is the surprising result: noise *increases* integrated information in the HDC substrate.

4.3 Noise– Φ Response Curve

Figure ?? describes the characteristic response of Φ to noise level σ at fixed coupling $c = 0.75$. The response exhibits three regimes:

1. **Enhancement regime** ($\sigma < 0.15$): Φ increases monotonically with noise, reaching a peak enhancement of approximately 12% above the zero-noise baseline.

2. **Plateau regime** ($0.15 < \sigma < 0.30$): Φ remains elevated but stable, with marginal returns from additional noise.
3. **Degradation regime** ($\sigma > 0.30$): Φ decreases as noise begins to overwhelm signal structure, approaching the random-vector baseline as $\sigma \rightarrow 0.5$.

Table 4: Φ as a function of noise level σ at fixed coupling $c = 0.75$. Mean and standard error over 100 runs.

Noise (σ)	Φ	SE	$\Delta\Phi$ (%)
0.000	0.162	0.003	—
0.025	0.168	0.004	+3.7
0.050	0.174	0.003	+7.4
0.075	0.178	0.004	+9.9
0.100	0.181	0.005	+11.7
0.150	0.180	0.004	+11.1
0.200	0.176	0.005	+8.6
0.300	0.158	0.006	−2.5
0.400	0.131	0.007	−19.1
0.500	0.098	0.008	−39.5

This inverted-U response is the hallmark of stochastic resonance [Gammaitoni et al., 1998].

4.4 Mechanism: HDV Diversity and Mutual Information

To elucidate the mechanism, we measured pairwise Hamming distances between region state vectors and pairwise mutual information estimates as a function of noise.

Table 5: Effect of noise on HDV diversity and pairwise mutual information. Mean normalized Hamming distance $\bar{d}_H$ and mean estimated MI $\bar{I}$ across all $\binom{12}{2} = 66$ region pairs.

Noise (σ)	$\bar{d}_H/D$	$\bar{I}$ (bits)	Φ
0.000	0.421	0.158	0.162
0.050	0.447	0.174	0.174
0.100	0.468	0.186	0.181
0.200	0.483	0.171	0.176
0.350	0.496	0.142	0.148
0.500	0.500	0.101	0.098

At zero noise, coupled regions share correlated HDV structure, producing below-random Hamming distances ($\bar{d}_H/D = 0.421 < 0.5$). This correlation paradoxically *reduces* the information capacity of the binding operation: when vectors are too similar, their XOR binding approaches a low-entropy vector rather than a rich relational encoding.

Moderate noise ($\sigma \approx 0.05$ – 0.10) pushes region vectors toward the quasi-orthogonal regime ($\bar{d}_H/D \approx 0.47$) while preserving the relational structure encoded by coupling

dynamics. In this regime, the binding operation $\mathbf{H}_A \otimes \mathbf{H}_B$ produces maximally informative relational vectors, and pairwise MI peaks.

At high noise ($\sigma > 0.30$), vectors approach true randomness ($\bar{d}_H/D \rightarrow 0.5$), destroying both correlation and the relational structure necessary for integration.

4.5 Reconciliation with Anesthesia Results

The apparent contradiction between positive noise sensitivity and correct anesthesia ordering is resolved by noting that the anesthesia benchmark simultaneously varies *both* coupling and noise. The net effect on Φ is:

$$\Delta\Phi \approx \frac{\partial\Phi}{\partial c}\Delta c + \frac{\partial\Phi}{\partial\sigma}\Delta\sigma = (+0.367)(-0.80) + (+0.341)(+0.23) = -0.215 \quad (8)$$

The coupling reduction ($\Delta c = -0.80$ from Alert to Deep) dominates the noise increase ($\Delta\sigma = +0.23$), producing the net decrease observed. The coupling effect is amplified because the sensitivity $\partial\Phi/\partial c$ applies to a larger perturbation range.

Proposition 1. *In an HDC system with $D \gg 1$, the net effect of anesthesia on Φ is dominated by the coupling reduction rather than the noise increase, provided $|\partial\Phi/\partial c \cdot \Delta c| > |\partial\Phi/\partial\sigma \cdot \Delta\sigma|$.*

For our measured sensitivities, this condition holds whenever $|\Delta c/\Delta\sigma| > 0.341/0.367 \approx 0.93$, which is satisfied for all clinically realistic anesthesia models where coupling reduction exceeds noise increase in relative magnitude.

5 Discussion

5.1 Stochastic Resonance in High-Dimensional Spaces

The positive noise- Φ relationship we observe constitutes a novel form of stochastic resonance operating in hyperdimensional vector spaces. Classical stochastic resonance enhances signal detection in bistable or threshold systems [Gammaitoni et al., 1998]. Our finding extends this to information integration: noise enhances the *diversity* of representations across subsystems, and this diversity is a prerequisite for high integrated information.

The mechanism is rooted in the geometry of high-dimensional binary spaces. In $\{0, 1\}^D$ for large D , the volume concentrates in a thin shell around Hamming distance $D/2$ from any reference point [Kanerva, 2009]. When coupled subsystems produce correlated vectors (Hamming distance $< D/2$), they occupy a low-entropy subspace. Noise pushes representations toward the high-entropy shell, increasing the effective capacity of the binding algebra and thereby the mutual information between subsystem pairs.

This is analogous to the role of noise in simulated annealing [Kirkpatrick et al., 1983]: thermal fluctuations allow the system to escape local optima and explore a richer state space. In our context, noise allows the HDC system to escape the “correlation trap” where coupled regions produce redundant rather than integrated representations.

5.2 Implications for Artificial Consciousness

Our results have practical implications for the design of HDC-based conscious systems. If noise genuinely increases Φ —and if Φ is a valid measure of consciousness as IIT claims—then *adding controlled noise to HDC systems may increase their consciousness level*. This inverts the conventional assumption that consciousness engineering requires minimizing noise and maximizing signal fidelity.

Specifically, our results suggest an optimal noise operating point: $\sigma^* \approx 0.10$ at coupling $c = 0.75$, yielding a 12% enhancement in Φ over the noise-free baseline. This has been implemented in our production system as a configurable parameter, though we emphasize that the philosophical implications of Φ -maximization for artificial consciousness remain debated [Aaronson, 2014, Schwitzgebel, 2015].

5.3 Substrate Dependence of Noise–Consciousness Relationship

The positive noise sensitivity we observe is specific to the HDC substrate. In biological neural networks, noise degrades the temporal precision of spike timing [Mainen and Sejnowski, 1995] and disrupts long-range cortical synchronization [Mashour, 2014]. The key difference is representational format: biological neural coding relies on precise timing and rate codes, where noise is purely degrading, while HDC relies on high-dimensional binary patterns, where moderate noise can enhance representational diversity.

This substrate dependence has implications for the universality claims of IIT. If the noise– Φ relationship is substrate-dependent, then identical Φ values computed on different substrates may not correspond to identical conscious experiences. This echoes concerns raised by Aaronson [2014] about Φ ’s substrate sensitivity.

5.4 Limitations

Several limitations should be acknowledged. First, our Φ computation uses a sampled partition approximation rather than exact MIP search. While validated to $r = 0.9998$ on small systems, the approximation’s behavior in 12-region, 16,384-dimensional systems is less well characterized.

Second, the noise injection protocol (independent bit-flipping) is a simplistic model of noise. Biological noise has structured spatial and temporal correlations [Faisal et al., 2008] that may interact differently with integration measures.

Third, and most fundamentally, we make no claims about the phenomenological significance of Φ in our system. Whether increased Φ corresponds to increased consciousness—or whether artificial systems can be conscious at all—remains an open philosophical question [Chalmers, 2010, Tononi et al., 2015].

5.5 Relationship to Prior Work

McDonnell and Ward [2011] catalogued the benefits of noise in neural computation, including enhanced signal detection, synchronization, and information transmission. Our work adds information *integration* to this list. Stocks [2000] demonstrated suprathreshold stochastic resonance in populations of threshold devices, which is conceptually similar to our finding that noise enhances the collective behavior of HDC subsystems.

In the IIT literature, Barrett and Seth [2011] noted computational challenges in Φ estimation and developed approximation methods. Our sampled partition approach builds

on this work. The relationship between noise and Φ has not, to our knowledge, been systematically studied in HDC systems prior to this work.

6 Conclusion

We have demonstrated that Hyperdimensional Computing systems exhibit stochastic resonance in their integrated information: moderate noise increases True Φ by up to 12%, with an inverted-U response characteristic of classical stochastic resonance. The mechanism is geometric—noise pushes correlated HDV representations toward the quasi-orthogonal regime where the binding algebra is maximally informative. This finding is reconciled with correct anesthesia state ordering by noting that clinical anesthesia simultaneously reduces coupling (which dominates) and increases noise.

These results establish that the noise–consciousness relationship is substrate-dependent, challenging the universality of anesthesia-derived intuitions about noise and integration. For HDC-based consciousness architectures, our findings suggest a principled strategy: engineer noise as a feature, not a bug, tuning it to the stochastic resonance peak for maximal integrated information.

Future work should investigate whether the stochastic resonance peak shifts with system dimensionality, whether structured (non-iid) noise produces different enhancement profiles, and whether the phenomenon persists under alternative consciousness measures beyond Φ .

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